

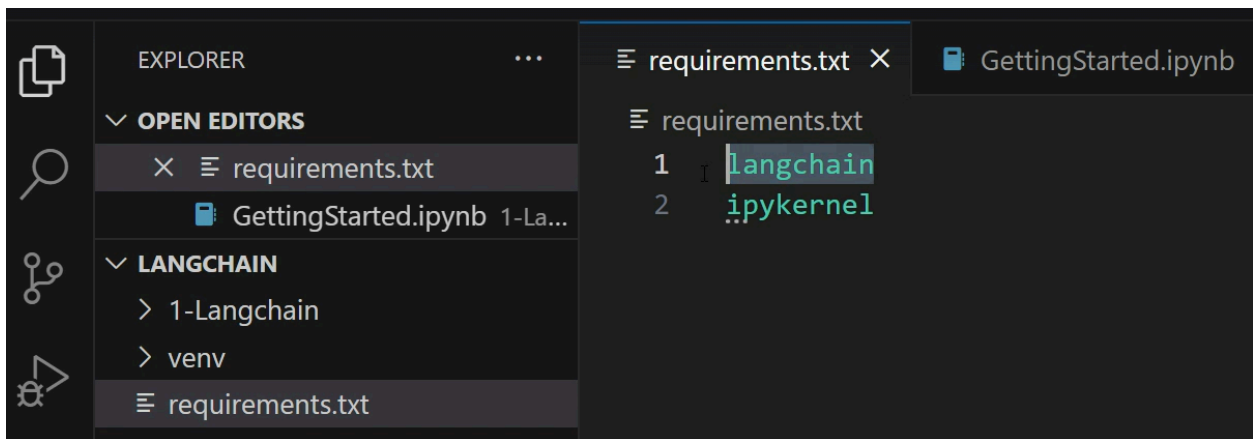
here we create a virtual environment for Langchain development in VS code

1. Create virtual environment

```
Microsoft Windows [Version 10.0.19045.4412]
(c) Microsoft Corporation. All rights reserved.

E:\UDemy Final\Langchain>conda create -p venv python==3.10 -y
WARNING: A space was detected in your requested environment path
'E:\UDemy Final\Langchain\venv'
Spaces in paths can sometimes be problematic.
Collecting package metadata (current_repodata.json): -
```

2. create requirement.txt and add langchain

The screenshot shows the VS Code interface. On the left, the Explorer sidebar is open, showing a file tree with folders '1-Langchain' and 'venv', and a file 'requirements.txt' selected. The main editor area shows the 'requirements.txt' file with two lines of code: 'langchain' and 'ipykernel'.

```
requirements.txt
1 langchain
2 ipykernel
```

3. activate Virtual environment use the below command.

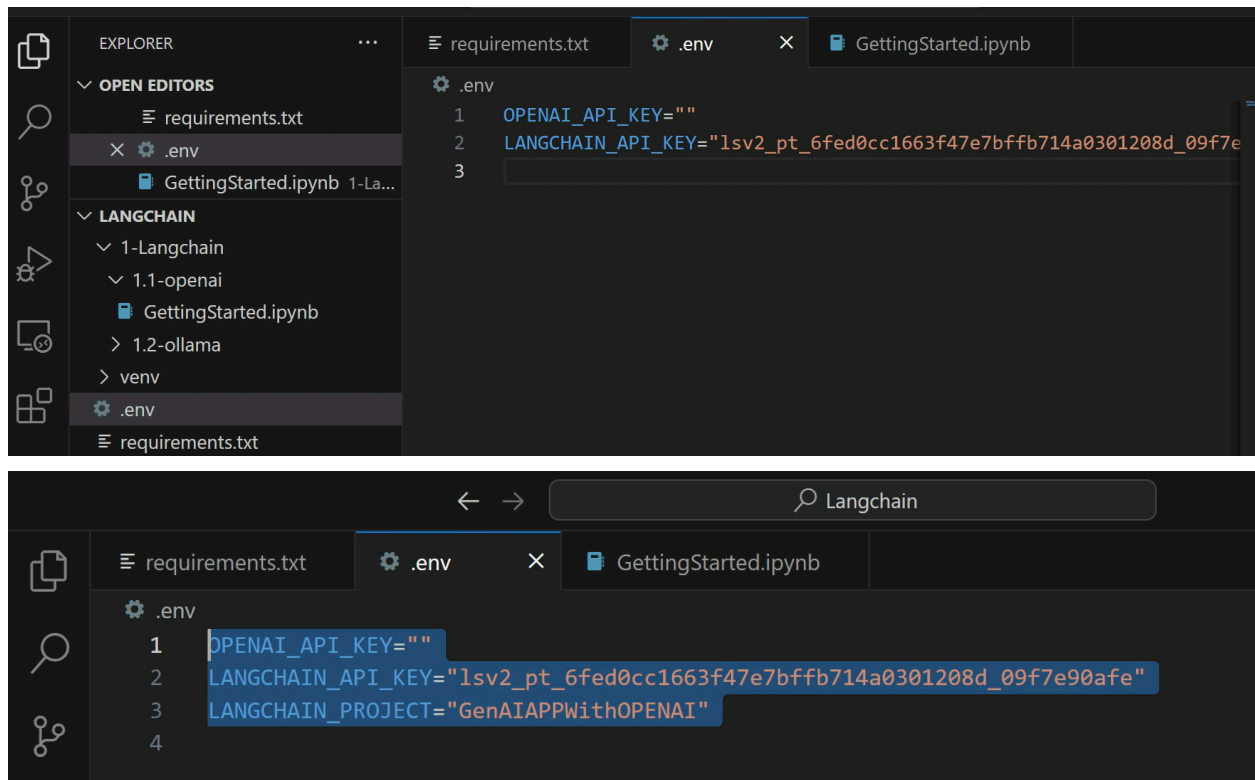
```
#
# To activate this environment, use
#
#     $ conda activate E:\UDemy Final\Langchain\venv
#
# To deactivate an active environment, use
#
#     $ conda deactivate

E:\UDemy Final\Langchain>conda activate venv/
```

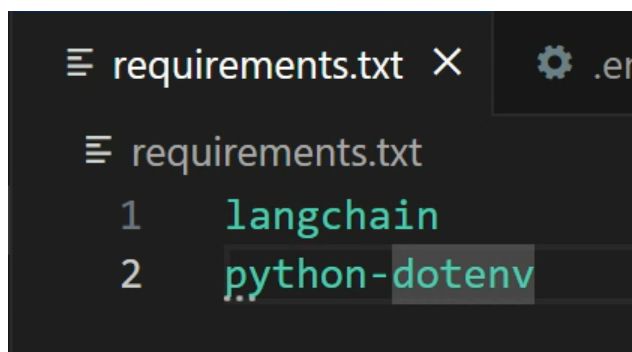
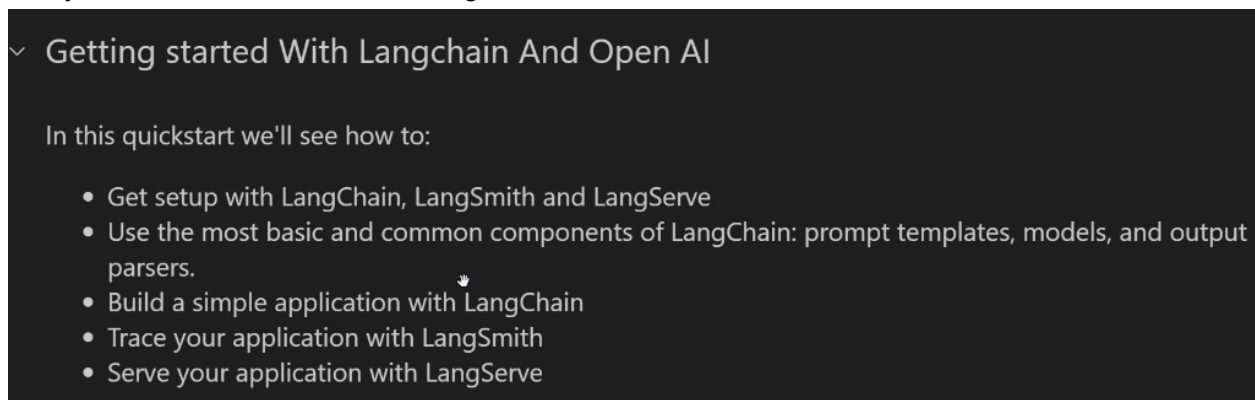
4. to install the packages mentioned in the requirements.txt use

```
(E:\UDemy Final\Langchain\venv) E:\UDemy Final\Langchain>pip insstall -r requirements.txt
```

5. Create .env file and add the api keys



finally this is what we will be learning



python-dotenv is a Python library that reads key-value pairs from a `.env` file and sets them as environment variables, helping developers follow the 12-factor app principles. It allows applications to load configuration from a `.env` file during development while remaining configurable via the actual environment, making it ideal for managing API keys, database credentials, and sensitive settings without hardcoding them.